

Data Cleaning using Pandas and Numpy

```
In [1]: import pandas as pd
```

```
In [2]: data=pd.read_csv("C:/Users/HP/Desktop/datasets/country.csv")
```

```
In [3]: data=pd.read_excel("C:/Users/HP/Downloads/COUNTRY.xlsx")
```

```
In [ ]:
```

renaming columns

`data.rename(columns={'old_name': 'new_name', 'B': 'new_B_name'}, inplace=True)`

```
In [4]: data.rename(columns={'Unnamed: 0': 'country', "GDP": "gdp"}, inplace=True)
```

```
In [5]: data
```

```
Out[5]:
```

	country	Population	gdp	Surface Area	HDI	Continent
0	canada	35.467	1785387	9984670	0.913	America
1	france	63.951	2833687	640679	0.888	Europe
2	italy	80.940	3874437	357114	0.916	Europe
3	germany	60.665	2167744	301336	0.873	Europe
4	japan	127.061	4602367	377930	0.891	Asia
5	uk	64.511	2950039	242495	0.907	Europe
6	us	318.523	17348075	9525067	0.915	America

```
In [6]: # Make column 'A' as the index  
data.set_index('country', inplace=True)
```

```
In [7]: data
```

Out[7]:

	Population	gdp	Surface Area	HDI	Continent
--	------------	-----	--------------	-----	-----------

country					
canada	35.467	1785387	9984670	0.913	America
france	63.951	2833687	640679	0.888	Europe
italy	80.940	3874437	357114	0.916	Europe
germany	60.665	2167744	301336	0.873	Europe
japan	127.061	4602367	377930	0.891	Asia
uk	64.511	2950039	242495	0.907	Europe
us	318.523	17348075	9525067	0.915	America

In [8]: `data.loc[["canada"]]`

Out[8]:

	Population	gdp	Surface Area	HDI	Continent
--	------------	-----	--------------	-----	-----------

country					
canada	35.467	1785387	9984670	0.913	America

In [9]: `data.loc[["canada", "uk", "us", "italy"]]`

Out[9]:

	Population	gdp	Surface Area	HDI	Continent
--	------------	-----	--------------	-----	-----------

country					
canada	35.467	1785387	9984670	0.913	America
uk	64.511	2950039	242495	0.907	Europe
us	318.523	17348075	9525067	0.915	America
italy	80.940	3874437	357114	0.916	Europe

In [10]: `data`

Out[10]:

	Population	gdp	Surface Area	HDI	Continent
--	------------	-----	--------------	-----	-----------

country					
canada	35.467	1785387	9984670	0.913	America
france	63.951	2833687	640679	0.888	Europe
italy	80.940	3874437	357114	0.916	Europe
germany	60.665	2167744	301336	0.873	Europe
japan	127.061	4602367	377930	0.891	Asia
uk	64.511	2950039	242495	0.907	Europe
us	318.523	17348075	9525067	0.915	America

In [11]: `data.iloc[6, -2] # rc`

Out[11]: 0.915

In []:

In [12]: data.iloc[0,-1]

Out[12]: 'America'

In [13]: *# Reset index to make 'A' appear as a column*
data.reset_index(inplace=True)

In [14]: data["gdp"]

Out[14]:

0	1785387
1	2833687
2	3874437
3	2167744
4	4602367
5	2950039
6	17348075

Name: gdp, dtype: int64

In [15]: NEW=data[["country","gdp","HDI"]]
NEW

Out[15]:

	country	gdp	HDI
0	canada	1785387	0.913
1	france	2833687	0.888
2	italy	3874437	0.916
3	germany	2167744	0.873
4	japan	4602367	0.891
5	uk	2950039	0.907
6	us	17348075	0.915

In []:

In [16]: data

```
Out[16]:
```

	country	Population	gdp	Surface Area	HDI	Continent
0	canada	35.467	1785387	9984670	0.913	America
1	france	63.951	2833687	640679	0.888	Europe
2	italy	80.940	3874437	357114	0.916	Europe
3	germany	60.665	2167744	301336	0.873	Europe
4	japan	127.061	4602367	377930	0.891	Asia
5	uk	64.511	2950039	242495	0.907	Europe
6	us	318.523	17348075	9525067	0.915	America

Conditional Selection

```
In [17]: new_data=data["gdp"]>3000000 #30 L
```

```
In [18]: new=data[data["Population"]>100]
```

```
In [19]: x=data["Population"]>100
data[x]
```

```
Out[19]:
```

	country	Population	gdp	Surface Area	HDI	Continent
4	japan	127.061	4602367	377930	0.891	Asia
6	us	318.523	17348075	9525067	0.915	America

```
In [20]: new_data
```

```
Out[20]:
```

0	False
1	False
2	True
3	False
4	True
5	False
6	True

Name: gdp, dtype: bool

```
In [21]: data #no change
```

```
Out[21]:
```

	country	Population	gdp	Surface Area	HDI	Continent
0	canada	35.467	1785387	9984670	0.913	America
1	france	63.951	2833687	640679	0.888	Europe
2	italy	80.940	3874437	357114	0.916	Europe
3	germany	60.665	2167744	301336	0.873	Europe
4	japan	127.061	4602367	377930	0.891	Asia
5	uk	64.511	2950039	242495	0.907	Europe
6	us	318.523	17348075	9525067	0.915	America

```
In [22]: europe=data[data["Continent"]=="Europe"]
```

```
In [23]: x=(data["Continent"]=="Europe") & (data["HDI"]>.9)
data[x]
```

```
Out[23]:
```

	country	Population	gdp	Surface Area	HDI	Continent
2	italy	80.940	3874437	357114	0.916	Europe
5	uk	64.511	2950039	242495	0.907	Europe

```
In [24]: europe_30 = data[(data["Continent"] == "Europe") & (data["HDI"] > 0.9)]
#When filtering a DataFrame based on multiple conditions, you need to use the operato
#| for "or", instead of the and and or keywords.
```

```
In [25]: europe_30
```

```
Out[25]:
```

	country	Population	gdp	Surface Area	HDI	Continent
2	italy	80.940	3874437	357114	0.916	Europe
5	uk	64.511	2950039	242495	0.907	Europe

```
In [26]: data
```

```
Out[26]:
```

	country	Population	gdp	Surface Area	HDI	Continent
0	canada	35.467	1785387	9984670	0.913	America
1	france	63.951	2833687	640679	0.888	Europe
2	italy	80.940	3874437	357114	0.916	Europe
3	germany	60.665	2167744	301336	0.873	Europe
4	japan	127.061	4602367	377930	0.891	Asia
5	uk	64.511	2950039	242495	0.907	Europe
6	us	318.523	17348075	9525067	0.915	America

```
In [27]: data.info()
```

```

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 7 entries, 0 to 6
Data columns (total 6 columns):
#   Column          Non-Null Count  Dtype
---  ---
0   country         7 non-null      object
1   Population       7 non-null      float64
2   gdp              7 non-null      int64
3   Surface Area    7 non-null      int64
4   HDI              7 non-null      float64
5   Continent        7 non-null      object
dtypes: float64(2), int64(2), object(2)
memory usage: 464.0+ bytes

```

In [28]: `data.describe()` *#quick statistical*

Out[28]:

	Population	gdp	Surface Area	HDI
count	7.000000	7.000000e+00	7.000000e+00	7.000000
mean	107.302571	5.080248e+06	3.061327e+06	0.900429
std	97.249970	5.494020e+06	4.576187e+06	0.016592
min	35.467000	1.785387e+06	2.424950e+05	0.873000
25%	62.308000	2.500716e+06	3.292250e+05	0.889500
50%	64.511000	2.950039e+06	3.779300e+05	0.907000
75%	104.000500	4.238402e+06	5.082873e+06	0.914000
max	318.523000	1.734808e+07	9.984670e+06	0.916000

In [29]: `data[["gdp", "Surface Area"]].describe()`

Out[29]:

	gdp	Surface Area
count	7.000000e+00	7.000000e+00
mean	5.080248e+06	3.061327e+06
std	5.494020e+06	4.576187e+06
min	1.785387e+06	2.424950e+05
25%	2.500716e+06	3.292250e+05
50%	2.950039e+06	3.779300e+05
75%	4.238402e+06	5.082873e+06
max	1.734808e+07	9.984670e+06

In [30]: `data`

```
Out[30]:
```

	country	Population	gdp	Surface Area	HDI	Continent
0	canada	35.467	1785387	9984670	0.913	America
1	france	63.951	2833687	640679	0.888	Europe
2	italy	80.940	3874437	357114	0.916	Europe
3	germany	60.665	2167744	301336	0.873	Europe
4	japan	127.061	4602367	377930	0.891	Asia
5	uk	64.511	2950039	242495	0.907	Europe
6	us	318.523	17348075	9525067	0.915	America

```
In [31]: data.head(3)
```

```
Out[31]:
```

	country	Population	gdp	Surface Area	HDI	Continent
0	canada	35.467	1785387	9984670	0.913	America
1	france	63.951	2833687	640679	0.888	Europe
2	italy	80.940	3874437	357114	0.916	Europe

```
In [32]: #data.select_dtypes()
```

```
In [33]: # Separate numerical and categorical columns
numerical_data = data.select_dtypes(include='number')
numerical_data
```

```
Out[33]:
```

	Population	gdp	Surface Area	HDI
0	35.467	1785387	9984670	0.913
1	63.951	2833687	640679	0.888
2	80.940	3874437	357114	0.916
3	60.665	2167744	301336	0.873
4	127.061	4602367	377930	0.891
5	64.511	2950039	242495	0.907
6	318.523	17348075	9525067	0.915

```
In [34]: categorical_data = data.select_dtypes(include='object')
categorical_data
```

Out[34]:

	country	Continent
0	canada	America
1	france	Europe
2	italy	Europe
3	germany	Europe
4	japan	Asia
5	uk	Europe
6	us	America

In [35]:

data

Out[35]:

	country	Population	gdp	Surface Area	HDI	Continent
0	canada	35.467	1785387	9984670	0.913	America
1	france	63.951	2833687	640679	0.888	Europe
2	italy	80.940	3874437	357114	0.916	Europe
3	germany	60.665	2167744	301336	0.873	Europe
4	japan	127.061	4602367	377930	0.891	Asia
5	uk	64.511	2950039	242495	0.907	Europe
6	us	318.523	17348075	9525067	0.915	America

In [36]:

data["Continent"].value_counts()

Out[36]:

Europe 4

America 2

Asia 1

Name: Continent, dtype: int64

In [37]:

data["Continent"].value_counts()

Out[37]:

Europe 4

America 2

Asia 1

Name: Continent, dtype: int64

In [38]:

#Sorting

data.sort_values(by='gdp') *#Ascending order*


```
Out[38]:
```

	country	Population	gdp	Surface Area	HDI	Continent
0	canada	35.467	1785387	9984670	0.913	America
3	germany	60.665	2167744	301336	0.873	Europe
1	france	63.951	2833687	640679	0.888	Europe
5	uk	64.511	2950039	242495	0.907	Europe
2	italy	80.940	3874437	357114	0.916	Europe
4	japan	127.061	4602367	377930	0.891	Asia
6	us	318.523	17348075	9525067	0.915	America

```
In [39]: data.sort_values(by='Population',ascending=False) #Descending order
```

```
Out[39]:
```

	country	Population	gdp	Surface Area	HDI	Continent
6	us	318.523	17348075	9525067	0.915	America
4	japan	127.061	4602367	377930	0.891	Asia
2	italy	80.940	3874437	357114	0.916	Europe
5	uk	64.511	2950039	242495	0.907	Europe
1	france	63.951	2833687	640679	0.888	Europe
3	germany	60.665	2167744	301336	0.873	Europe
0	canada	35.467	1785387	9984670	0.913	America

```
In [40]: data.dtypes
```

```
Out[40]: country      object
Population  float64
gdp         int64
Surface Area int64
HDI         float64
Continent   object
dtype: object
```

```
In [41]: data.shape #RC
```

```
Out[41]: (7, 6)
```

```
In [42]: data.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 7 entries, 0 to 6
Data columns (total 6 columns):
#   Column          Non-Null Count  Dtype
---  ---
0   country         7 non-null      object
1   Population       7 non-null      float64
2   gdp              7 non-null      int64
3   Surface Area    7 non-null      int64
4   HDI              7 non-null      float64
5   Continent        7 non-null      object
dtypes: float64(2), int64(2), object(2)
memory usage: 464.0+ bytes
```

```
In [43]: #Dropping rows
data.drop([0,2,5]) #Won't reflect in data
```

```
Out[43]:
```

	country	Population	gdp	Surface Area	HDI	Continent
1	france	63.951	2833687	640679	0.888	Europe
3	germany	60.665	2167744	301336	0.873	Europe
4	japan	127.061	4602367	377930	0.891	Asia
6	us	318.523	17348075	9525067	0.915	America

```
In [44]: x=data["Continent"]!="Europe"
data[x]
```

```
Out[44]:
```

	country	Population	gdp	Surface Area	HDI	Continent
0	canada	35.467	1785387	9984670	0.913	America
4	japan	127.061	4602367	377930	0.891	Asia
6	us	318.523	17348075	9525067	0.915	America

```
In [45]: data
```

```
Out[45]:
```

	country	Population	gdp	Surface Area	HDI	Continent
0	canada	35.467	1785387	9984670	0.913	America
1	france	63.951	2833687	640679	0.888	Europe
2	italy	80.940	3874437	357114	0.916	Europe
3	germany	60.665	2167744	301336	0.873	Europe
4	japan	127.061	4602367	377930	0.891	Asia
5	uk	64.511	2950039	242495	0.907	Europe
6	us	318.523	17348075	9525067	0.915	America

```
In [ ]:
```

```
In [46]: df = data[data['Continent'] != 'Europe']
```

```
df
```

```
Out[46]:
```

	country	Population	gdp	Surface Area	HDI	Continent
0	canada	35.467	1785387	9984670	0.913	America
4	japan	127.061	4602367	377930	0.891	Asia
6	us	318.523	17348075	9525067	0.915	America

```
In [47]: #Dropping single column  
data.drop(columns='Population')
```

```
Out[47]:
```

	country	gdp	Surface Area	HDI	Continent
0	canada	1785387	9984670	0.913	America
1	france	2833687	640679	0.888	Europe
2	italy	3874437	357114	0.916	Europe
3	germany	2167744	301336	0.873	Europe
4	japan	4602367	377930	0.891	Asia
5	uk	2950039	242495	0.907	Europe
6	us	17348075	9525067	0.915	America

```
In [48]: #Dropping multiple column  
data.drop(columns=['Population', 'HDI'])
```

```
Out[48]:
```

	country	gdp	Surface Area	Continent
0	canada	1785387	9984670	America
1	france	2833687	640679	Europe
2	italy	3874437	357114	Europe
3	germany	2167744	301336	Europe
4	japan	4602367	377930	Asia
5	uk	2950039	242495	Europe
6	us	17348075	9525067	America

CREATING NEW COLUMNS

```
In [49]: data
```

Out[49]:

	country	Population	gdp	Surface Area	HDI	Continent
0	canada	35.467	1785387	9984670	0.913	America
1	france	63.951	2833687	640679	0.888	Europe
2	italy	80.940	3874437	357114	0.916	Europe
3	germany	60.665	2167744	301336	0.873	Europe
4	japan	127.061	4602367	377930	0.891	Asia
5	uk	64.511	2950039	242495	0.907	Europe
6	us	318.523	17348075	9525067	0.915	America

In [50]: `data['GDP_times_Population'] = data['gdp'] * data['Population']`

In [51]: `data`

Out[51]:

	country	Population	gdp	Surface Area	HDI	Continent	GDP_times_Population
0	canada	35.467	1785387	9984670	0.913	America	6.332232e+07
1	france	63.951	2833687	640679	0.888	Europe	1.812171e+08
2	italy	80.940	3874437	357114	0.916	Europe	3.135969e+08
3	germany	60.665	2167744	301336	0.873	Europe	1.315062e+08
4	japan	127.061	4602367	377930	0.891	Asia	5.847814e+08
5	uk	64.511	2950039	242495	0.907	Europe	1.903100e+08
6	us	318.523	17348075	9525067	0.915	America	5.525761e+09

In [52]: `#Calculating GDP per capita=GDP/Population`
`data["per capita"]=data['gdp']/data['Population']`

In [53]: `data`

Out[53]:

	country	Population	gdp	Surface Area	HDI	Continent	GDP_times_Population	per capita
0	canada	35.467	1785387	9984670	0.913	America	6.332232e+07	50339.385908
1	france	63.951	2833687	640679	0.888	Europe	1.812171e+08	44310.284437
2	italy	80.940	3874437	357114	0.916	Europe	3.135969e+08	47868.013343
3	germany	60.665	2167744	301336	0.873	Europe	1.315062e+08	35733.025633
4	japan	127.061	4602367	377930	0.891	Asia	5.847814e+08	36221.712406
5	uk	64.511	2950039	242495	0.907	Europe	1.903100e+08	45729.239975
6	us	318.523	17348075	9525067	0.915	America	5.525761e+09	54464.120330

In [54]: `data.drop(columns=["HDI", "gdp"], inplace=True)`

```
In [55]: data
```

```
Out[55]:
```

	country	Population	Surface Area	Continent	GDP_times_Population	per capita
0	canada	35.467	9984670	America	6.332232e+07	50339.385908
1	france	63.951	640679	Europe	1.812171e+08	44310.284437
2	italy	80.940	357114	Europe	3.135969e+08	47868.013343
3	germany	60.665	301336	Europe	1.315062e+08	35733.025633
4	japan	127.061	377930	Asia	5.847814e+08	36221.712406
5	uk	64.511	242495	Europe	1.903100e+08	45729.239975
6	us	318.523	9525067	America	5.525761e+09	54464.120330

```
In [56]: langs=pd.Series(['French','German','Italian'],  
                        index=[1,2,3],  
                        name='Language')  
langs
```

```
Out[56]:
```

1	French
2	German
3	Italian

Name: Language, dtype: object

```
In [57]: data["Language"]=langs
```

```
In [58]: data
```

```
Out[58]:
```

	country	Population	Surface Area	Continent	GDP_times_Population	per capita	Language
0	canada	35.467	9984670	America	6.332232e+07	50339.385908	NaN
1	france	63.951	640679	Europe	1.812171e+08	44310.284437	French
2	italy	80.940	357114	Europe	3.135969e+08	47868.013343	German
3	germany	60.665	301336	Europe	1.315062e+08	35733.025633	Italian
4	japan	127.061	377930	Asia	5.847814e+08	36221.712406	NaN
5	uk	64.511	242495	Europe	1.903100e+08	45729.239975	NaN
6	us	318.523	9525067	America	5.525761e+09	54464.120330	NaN

```
In [59]: data.isnull().sum()
```

```
Out[59]:
```

country	0
Population	0
Surface Area	0
Continent	0
GDP_times_Population	0
per capita	0
Language	4

dtype: int64

```
In [60]: data.dropna()
```

```
Out[60]:
```

	country	Population	Surface Area	Continent	GDP_times_Population	per capita	Language
1	france	63.951	640679	Europe	1.812171e+08	44310.284437	French
2	italy	80.940	357114	Europe	3.135969e+08	47868.013343	German
3	germany	60.665	301336	Europe	1.315062e+08	35733.025633	Italian

```
In [61]: data
```

```
Out[61]:
```

	country	Population	Surface Area	Continent	GDP_times_Population	per capita	Language
0	canada	35.467	9984670	America	6.332232e+07	50339.385908	NaN
1	france	63.951	640679	Europe	1.812171e+08	44310.284437	French
2	italy	80.940	357114	Europe	3.135969e+08	47868.013343	German
3	germany	60.665	301336	Europe	1.315062e+08	35733.025633	Italian
4	japan	127.061	377930	Asia	5.847814e+08	36221.712406	NaN
5	uk	64.511	242495	Europe	1.903100e+08	45729.239975	NaN
6	us	318.523	9525067	America	5.525761e+09	54464.120330	NaN

```
In [62]: data.fillna("not avilable")
```

```
Out[62]:
```

	country	Population	Surface Area	Continent	GDP_times_Population	per capita	Language
0	canada	35.467	9984670	America	6.332232e+07	50339.385908	not avilable
1	france	63.951	640679	Europe	1.812171e+08	44310.284437	French
2	italy	80.940	357114	Europe	3.135969e+08	47868.013343	German
3	germany	60.665	301336	Europe	1.315062e+08	35733.025633	Italian
4	japan	127.061	377930	Asia	5.847814e+08	36221.712406	not avilable
5	uk	64.511	242495	Europe	1.903100e+08	45729.239975	not avilable
6	us	318.523	9525067	America	5.525761e+09	54464.120330	not avilable

```
In [63]: #replacing values per column

data['Language']=data['Language'].replace('French','English')
#occurences of french replaced to english
data
```

Out[63]:

	country	Population	Surface Area	Continent	GDP_times_Population	per capita	Language
0	canada	35.467	9984670	America	6.332232e+07	50339.385908	NaN
1	france	63.951	640679	Europe	1.812171e+08	44310.284437	English
2	italy	80.940	357114	Europe	3.135969e+08	47868.013343	German
3	germany	60.665	301336	Europe	1.315062e+08	35733.025633	Italian
4	japan	127.061	377930	Asia	5.847814e+08	36221.712406	NaN
5	uk	64.511	242495	Europe	1.903100e+08	45729.239975	NaN
6	us	318.523	9525067	America	5.525761e+09	54464.120330	NaN

In [64]:

```
# Define dictionary of values to replace
replace_dict = {"Europe": "EU", "America": "USA"}

# Replace values in the 'Continent' column
data['Continent'] = data['Continent'].replace(replace_dict)

data
```

Out[64]:

	country	Population	Surface Area	Continent	GDP_times_Population	per capita	Language
0	canada	35.467	9984670	USA	6.332232e+07	50339.385908	NaN
1	france	63.951	640679	EU	1.812171e+08	44310.284437	English
2	italy	80.940	357114	EU	3.135969e+08	47868.013343	German
3	germany	60.665	301336	EU	1.315062e+08	35733.025633	Italian
4	japan	127.061	377930	Asia	5.847814e+08	36221.712406	NaN
5	uk	64.511	242495	EU	1.903100e+08	45729.239975	NaN
6	us	318.523	9525067	USA	5.525761e+09	54464.120330	NaN

In [65]:

```
# Update the row with index label 4 to have values of China
data.loc[4] = ["INDIA", 1444216107, 14342903, "eu", 9596961, 0.758, "Asia"]

data
```

Out[65]:

	country	Population	Surface Area	Continent	GDP_times_Population	per capita	Language
0	canada	3.546700e+01	9984670	USA	6.332232e+07	50339.385908	NaN
1	france	6.395100e+01	640679	EU	1.812171e+08	44310.284437	English
2	italy	8.094000e+01	357114	EU	3.135969e+08	47868.013343	German
3	germany	6.066500e+01	301336	EU	1.315062e+08	35733.025633	Italian
4	INDIA	1.444216e+09	14342903	eu	9.596961e+06	0.758000	Asia
5	uk	6.451100e+01	242495	EU	1.903100e+08	45729.239975	NaN
6	us	3.185230e+02	9525067	USA	5.525761e+09	54464.120330	NaN

